1.Comparison of Software Architecture Patterns

For the RP-Cloud Labs Tool-Session Service, the three architecture patterns can be compared using four criteria: typical scale, deployment, data ownership, and operational complexity.

Comparison criteria	Client-Server	MVC (Model-View-Controller)	Micro-services
Typical scale	Small to large applications with clients communicating with a central server.	Small to medium applications, especially web applications.	Medium to very large applications with multiple independent services.
Deployment	The client and server are deployed separately.	Usually deployed as one application containing Model, View, and Controller components.	Each service can be deployed independently.
Data ownership	Data is usually managed centrally by the server.	The Model manages application data and business logic.	Each micro-service can own and manage its own database.
Operational complexity	Low to moderate; relatively easy to develop and manage.	Low to moderate; simple to organize and maintain.	High; requires service communication, monitoring, orchestration, and distributed management.

2. Recommended Architecture: Micro-services

Recommended for RP-Cloud Labs

Our group chooses Micro-services Architecture for the RP-Cloud Labs Tool-Session Service because it allows the system to be divided into smaller, independent services, making it easier to develop, test, and maintain. It supports separation of concerns because each service handles a specific responsibility, such as session management, account validation, and licence management. It also promotes low coupling, allowing services to communicate through APIs and events without depending heavily on each other's internal implementation. Although microservices have higher operational complexity, they provide flexibility, scalability, and independent deployment as more colleges join RP-Cloud Labs.